

# Ron Medina

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Machine Learning Engineer with a mathematics background. I build and ship ML and deep learning systems that optimize operations and surface useful insight — with attention to the infrastructure and engineering that keeps them running in production.

## Employment History

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### Sr. Machine Learning Engineer | Afni

Jan 2025 – Present

- Designed a repeatable feature engineering, preprocessing, training, and monitoring pipeline with a clean deployment interface for ML models served via Azure Functions; trained per-program per-KPI models, overcoming sparsity through extensive EDA
- A/B tested the hiring platform against the legacy process — hires made via the intelligence platform showed a 6% increase in lifetime over the same time period, alongside significant improvement in KPI performance
- Built predictive models for customer and employee retention risk, and ranking/recommender-style models for employee performance across program placements
- Models above serve as the core backend to the hiring platform that evaluates 5,000+ monthly candidates across all PH sites
- Designed a document analysis pipeline processing up to 30K documents daily with failure tracking, retries, and sufficient throughput; documents are attributed to the organizational hierarchy (employee, coach, manager) supporting RBAC, and tagged using NLP and LLM tools
- Built a monitoring and analytics dashboard with role-based access — users only see analytical results for documents they have access to based on their org hierarchy
- Implemented an agentic solution for determining ideal program placement of candidate hires, combining program description/LOB, contractual obligations, headcount fill requirements, candidate history, and interview/assessment scores
- Collaborated with multiple VPs on feature discussions, roadmap, capabilities, and communications
- Direct report to the VP of Software Engineering & AI on day-to-day operations and progress
- Managed a team of 4 vendors/contractors and led 2 in-house developers
- Ran POCs with other teams for the AI Team

### Machine Learning Engineer | Ubiquity

Aug 2022 – Jan 2025

- R&D / creating POCs and MVPs of AI products applicable to client domain
- Perform evaluation, cost-estimation, deployment strategies based on infrastructure constraints and business goals
- Developed and deployed transcription service which transcribed 1.2M+ production calls
- Reduced cost by 6X and increased accuracy of speech recognition system by +10%
- Developed and designed a fault tolerant distributed offline task queue service to scale speech recognition
- Helped develop the backend application for searching and filtering transcriptions
- Developed AI services for downstream processing, modeling, and analytics of call transcripts
- Extended existing open-source annotation tool for human data labeling to serve our internal use-case
- Developed a Transformer-based semantic search engine that runs performant on CPU
- Helped develop a masking model for images (screenshots) containing sensitive data using Tesseract
- Within 3 months after hiring, presented a POC for extending call recorder system from mono to stereo recording using socket programming and Avaya APIs which became the basis for a major project for the Telecommunications team.

### Junior Data Scientist | Sitel

June 2021 – Aug 2022

- Works directly under the Director of District Operations Quality Management
- Design of KPIs and metrics for various business processes
- Creation and deployment of Power BI dashboards
- Creation and deployment of APIs for integrating ML algorithms with existing products
- Ensure integrity and accuracy of reports

### Machine Learning Engineer | BrewedLogic, Inc.

Mar 2020 – June 2021

- Preprocessing, analysis, and feature engineering of datasets (1M-5M) for model training, selection, and evaluation
- Research and implementation of novel techniques and architectures for RecSys and fraud detection
- Developed RecSys used by Crisp restaurant platform and deployed on 26 franchises each with multiple stores in the US
- The collaborative filtering based model was written using NumPy, Scikit-Learn, and Pandas, and served in a Django server. Deployment of food RecSys increased average ticket count from 3.38 to 4.86 and average ticket value from \$11.49 to \$14.21 after the first two months in production.
- Worked with a senior data scientist on customer segmentation and sales forecasting

- Worked part-time at an E-commerce shop with hundreds of daily transactions.
- Performed basic statistical analysis of customer data including clustering based on RFM criteria.
- Analysis of frequently bought together items using graphs, Markov chains, and correlations.
- Created reports about product ROI rate.
- Helped grew follower count from 30,000 to 100,000+.

## Skills

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### Data Analysis

- Data visualization, data wrangling, and EDA using Pandas, seaborn, matplotlib, and NumPy.
- SQL, probability modelling, statistics, clustering

### Machine Learning

- Deep neural networks in TensorFlow and PyTorch
- Machine learning models in scikit-learn
- Recommender systems, anomaly detection / imbalanced learning
- Weak supervision for training noise-aware models
- Gradient Boosting (Catboost, XGBoost, LightGBM), ensembling/stacking

### Model Deployment and MLOps

- Development and deployment of prediction APIs (Django, FastAPI, Flask)
- Automation: Gitlab CI/CD, pdm, GitHub Actions, tox, Makefiles
- Task management with Celery
- MLflow experiment tracking, system metrics, evaluation, and model management
- Production quality Python packages with pdm
- Version control using git
- Unit testing with pytest, differential, and regression testing
- Containerization with Docker
- AWS Lambda, SQS and RabbitMQ, S3, RDS, EC2 / Auto Scaling groups
- PostgreSQL, MySQL, and Redis

### Algorithms and Data Structures

- Gold level in Problem Solving and Python @ Hackerrank

### Writing

- Author of OK Transformer — a collection of notebooks and articles on deep learning, ML engineering, and MLOps. Auto build/deploy via GitHub Actions + tox. Featured in the Gallery of Jupyter Books.
- Contributed to the mathematics appendix of Dive into Deep Learning; acknowledged as a contributor.

## Education

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**University of the Philippines - Diliman**

**06/13 – 12/18 (courses), 09/23 – 01/24 (thesis)**

*Bachelor of Science*

Major in Mathematics

**Awards:** University Scholar, 2nd Semester 2013-2014. GWA: 1.23

**Thesis:** *An Introduction to Finite Frames and a QR Factorization Approach for Constructing MB Frames*

**Relevant courses:** Intro to Computer Science (Python), Numerical Analysis, General Relativity (MS / PhD level), Linear Algebra, Advanced Calculus

**Eskwelabs**

**July 2019 – Oct 2019**

*Data Science Bootcamp*

Attended a 10-week bootcamp which included 160 hours of in-class learning in addition to coursework. At the end of the bootcamp, I presented my capstone project about modeling nonlinear chaotic systems using neural networks implemented in TensorFlow 1.x to industry leaders in Makati City, Philippines.